

PROJECT TRAINING REPORT

**Domain: Metagenomics And Gut Microbiome
Analysis**

A report on the project title

**“ML Based Analysis of Gut Microbiome Diversity in
Healthy vs. Disease”**

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Gut Microbiome Diversity: Healthy vs IBD Patients(ML based approach)

1. Objective

The primary objective of this project was to investigate differences in gut microbiome diversity between healthy individuals and patients diagnosed with Inflammatory Bowel Disease (IBD) and to develop machine learning models capable of accurately classifying disease status based on microbial abundance profiles and diversity biomarkers.

The clinical significance of this work lies in the growing evidence that gut microbial dysbiosis plays a central role in IBD pathogenesis. Identifying microbial biomarkers associated with disease status may contribute to the development of non-invasive diagnostic tools and support personalized microbiome-based therapeutic strategies.

2. Dataset Description & Preprocessing

2.1 Dataset Characteristics

The final microbiome dataset contained stool microbiome profiles collected from healthy individuals and IBD patients.

2.2 Dataset Summary

Dataset	Samples	Features
Full Dataset	1374	79
Training Dataset	1235	79
Testing Dataset	138	79

Table 1: Dataset summary

2.3 Class Distribution (Training Data)

Disease Status	Count
IBD (0)	920
Healthy (1)	315

Table 2: Class Distribution of Training Data

The dataset was moderately imbalanced, with IBD samples representing approximately 74.5% of the training dataset.

3. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

1. Duplicate label columns were removed to prevent data leakage.
2. Numerical microbiome abundance features were isolated for downstream analysis.
3. Features containing more than 90% zero values were evaluated for removal.
4. Missing values were imputed using the preprocessing pipeline implemented in the notebook.
5. Z-score standardization was applied to all microbial abundance features
Z-score standardization is a normalization technique that transforms features to have a mean of 0 and a standard deviation of 1.
6. The same preprocessing workflow was applied to both training and testing datasets.

Post-Processing Dataset Shape

Parameter	Value
Original Features	76
Remaining Features After Filtering	76
Scaled Feature Matrix Shape	(1235, 76)
Target Shape	(1235,)

Table 3: Post-Processing Dataset Shape

Target Variable

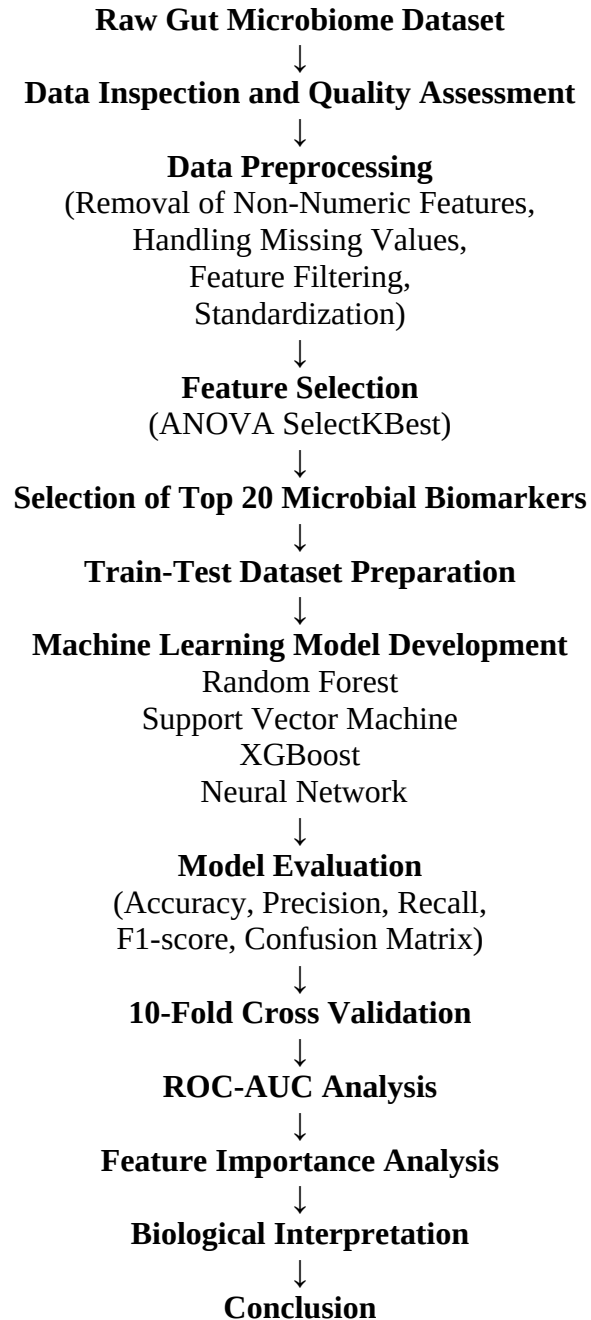
Disease status was encoded as:

Label	Class
0	IBD
1	Healthy

Table 4: Target Variable Encoding

The Disease column was retained as the binary target variable throughout model training and evaluation.

4. Workflow & Methodology



5. Exploratory Data Analysis (EDA)

Initial exploratory analysis included:

- Dataset shape inspection
- Feature distribution assessment
- Class distribution analysis
- Verification of preprocessing outputs
- Visualization of model performance through confusion matrices, ROC curves, feature importance plots, and cross-validation performance graphs

Feature Selection

Feature selection is the process of identifying the most informative variables for classification while removing irrelevant or redundant features.

The original microbiome dataset contained 76 microbial abundance features. Using all features may introduce noise and reduce model interpretability. ANOVA SelectKBest was therefore applied to identify the most discriminative biomarkers associated with IBD status. The top 20 features were selected based on their statistical ability to distinguish Healthy and IBD groups. This approach improved biological interpretability while reducing dimensionality and computational complexity.

SelectKBest (f_classif)

This approach statistically evaluates each microbial feature and assigns an F-score representing its ability to distinguish between Healthy and IBD groups.

The top 20 highest-scoring features were retained for machine learning analysis.

The top 20 features were selected because they represented the strongest statistical signals in the dataset while maintaining biological interpretability. This reduced the dimensionality of the microbiome dataset by approximately 74% (76 → 20 features) while retaining the most informative microbial biomarkers.

Data Splitting

The dataset was split into:

Dataset	Samples
Training Set	1235
Testing Set	138

Table 5: Training and Testing Dataset Split

The test dataset represented approximately 10% of the total dataset

6. Machine Learning Models

Four machine learning models were trained using the selected microbial biomarkers:

1. Random Forest
 2. Support Vector Machine (SVM)
 3. XGBoost
 4. Feed-Forward Neural Network
- Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification performance.
 - Support Vector Machine is a supervised learning algorithm that identifies the optimal hyperplane separating classes.
 - XGBoost is a gradient boosting algorithm that sequentially builds decision trees to correct previous prediction errors.
 - A Neural Network is a deep learning architecture composed of interconnected neurons organized into layers.

Additionally:

- 10-fold Cross Validation was applied to Random Forest, SVM, and XGBoost models.
- ROC-AUC analysis was performed for all four models

7. Outputs & Performance Metrics

7.1 Selected Microbial Biomarkers

The top 20 features identified through ANOVA feature selection were:

Rank	Feature	ANOVA Score
1	Gemmiger formicilis	106.35
2	observed_richness	106.08
3	Alistipes shahii	86.17
4	Barnesiella intestinihominis	75.91
5	shannon_diversity	65.99
6	Firmicutes bacterium CAG 83	60.80
7	Alistipes putredinis	56.27
8	Ruminococcus bromii	56.13
9	Oscillibacter sp 57 20	55.23
10	Eubacterium siraeum	46.65
11	fecalcal	44.91
12	Eubacterium ventriosum	43.83
13	Ruminococcus lactaris	41.75

14	Akkermansia muciniphila	40.81
15	Bacteroides fragilis	36.37
16	Flavonifractor plautii	33.57
17	Bifidobacterium adolescentis	33.27
18	Agathobaculum butyriciproducens	31.95
19	Ruminococcus bicirculans	30.65
20	Clostridium symbiosum	28.16

Table: 6 Top 20 Selected Microbial Biomarkers and ANOVA Scores

7.2 Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	Mean CV Accuracy
Random Forest	0.9275	0.9310	0.7714	0.8438	0.9102
SVM	0.9058	0.7895	0.8571	0.8219	0.8721
XGBoost	0.9420	0.9355	0.8286	0.8788	0.8980
Neural Network	0.8768	0.8214	0.6571	0.7302	0.8500

Table 7: Machine Learning Model Comparison

Model	Strength
Random Forest	Most interpretable
SVM	Highest Recall (0.8571)
XGBoost	Highest Accuracy and AUC
Neural Network	Deep learning benchmark

Table 8: Model Strength Comparison

7.3 ROC-AUC Analysis

AUC Scores

Model	AUC
Random Forest	0.9745
SVM	0.9592
XGBoost	0.9803
Neural Network	0.9173

Table 9: ROC-AUC Analysis Results

XGBoost achieved the highest overall performance with Accuracy = 94.20% and AUC = 0.9803, making it the most effective classifier for microbiome-based IBD prediction. SVM achieved the highest recall, indicating stronger sensitivity for identifying Healthy samples. Random Forest provided excellent interpretability through feature importance analysis, while the Neural Network served as a deep learning benchmark but showed lower performance after dimensionality reduction.

7.4 Confusion Matrix Summary

Random Forest

Metric	Value
TN	101
FP	2
FN	8
TP	27

SVM

Metric	Value
TN	95
FP	8
FN	5
TP	30

XGBoost

Metric	Value
TN	101
FP	2
FN	6
TP	29

Neural Network

Metric	Value
TN	98
FP	5
FN	12
TP	23

Table 10: Confusion Matrix Summary

Visual Output Interpretation

7.5 Feature Selection Plot

The feature-selection plot ranks microbial taxa according to ANOVA F-scores. Higher scores indicate stronger discriminatory power between Healthy and IBD groups.

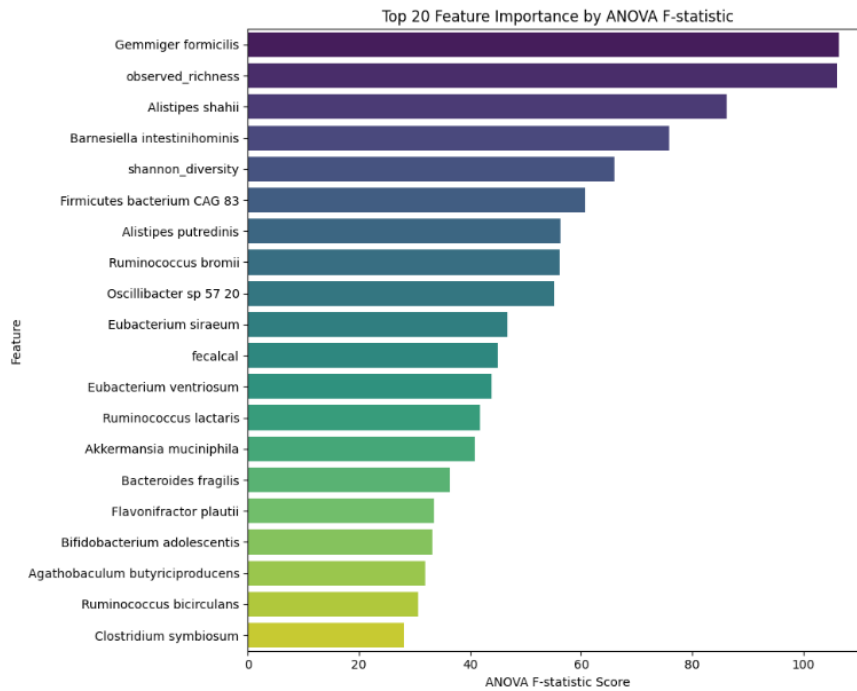


Fig 1: Feature Selection Plot (Top 20 Microbial Biomarkers)

7.6 Feature Importance Plot

The feature importance graph generated from the Random Forest model quantifies the contribution of each selected microbial biomarker to disease classification.

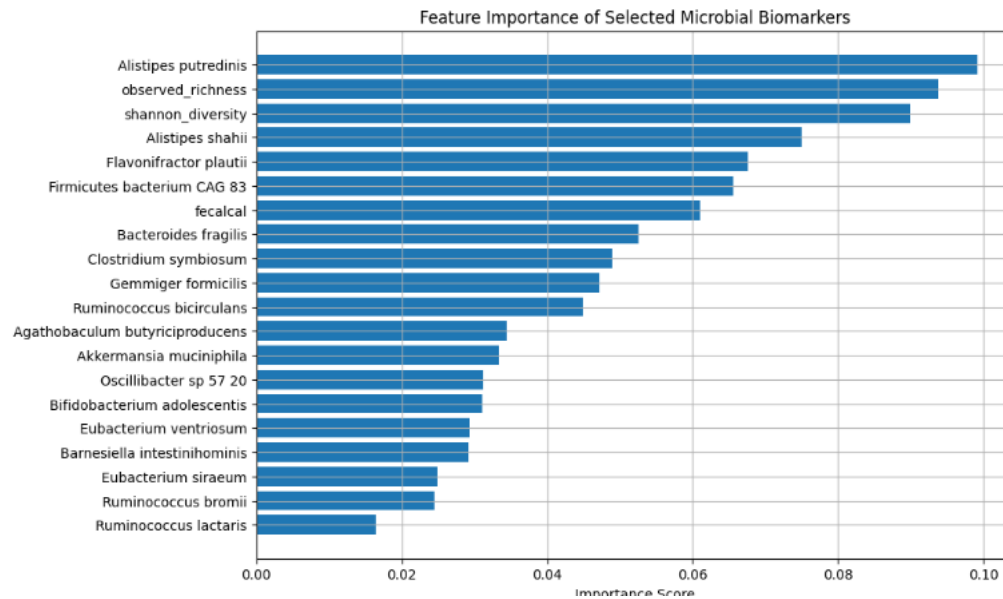


Fig 2: Random Forest Feature Importance Plot

7.7 ROC Curve

The ROC curve plots: Curves closer to the upper-left corner indicate stronger classification performance.

- X-axis: False Positive Rate (FPR)
- Y-axis: True Positive Rate (TPR)

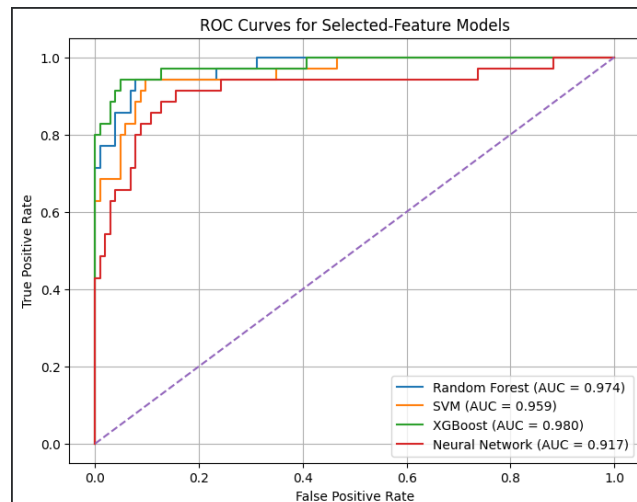


Fig 3: ROC Curve Comparison of Machine Learning Models

7.8 Confusion Matrix

Confusion matrices summarize model predictions by displaying:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

These matrices provide insight into model sensitivity and specificity.

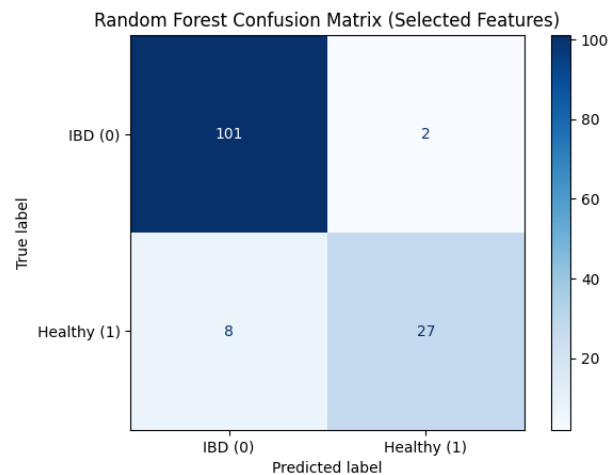


Fig 4: Confusion matrix- Random Forest

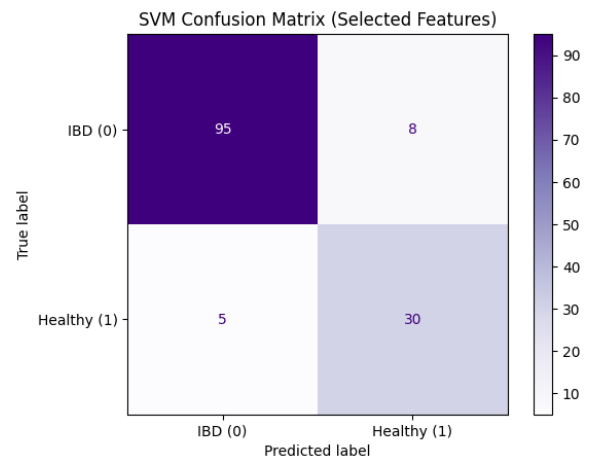


Fig 5: Confusion matrix- Support Vector Machine (SVM)

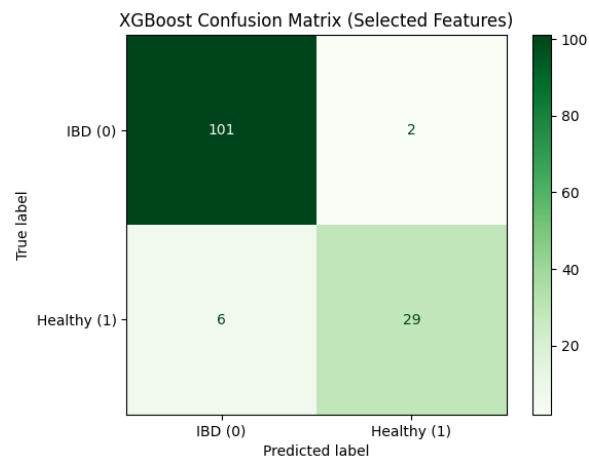


Fig 6: Confusion matrix- XGBoost

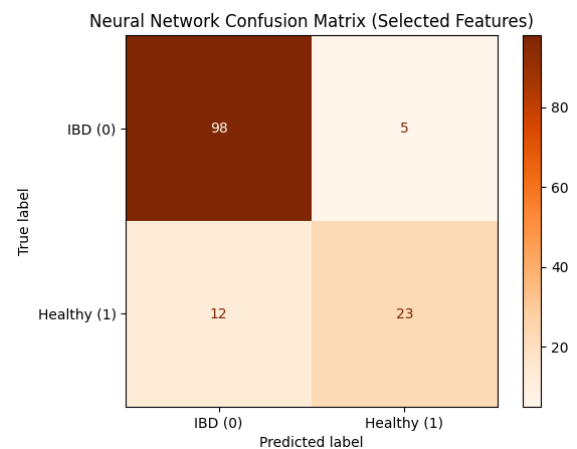


Fig 7: Confusion matrix- Neural Network

7.9 Model performance comparison:

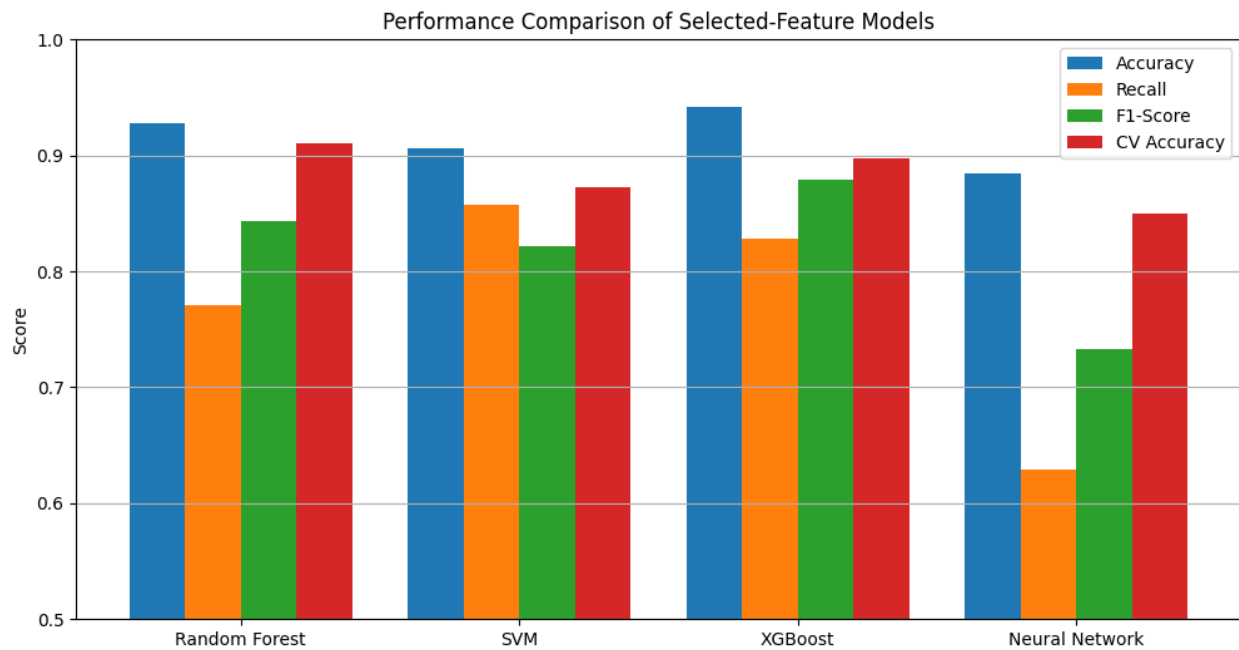


Fig 8: Model Performance Comparison Bar Plot

The model comparison bar plot provides a visual summary of the performance of the four machine learning models: Random forest, SVM, XGBoost and neural network. It compares key metrics such as accuracy, recall, F1 score and mean cross-validation accuracy. Generally, XGboost and random forest models show strong performance across all metrics, with XGBoost slightly outperforming in overall accuracy and F1 score, while the neural network model exhibits comparatively lower score in this specific comparison of selected features.

8. Result

8.1 Reduced Microbial Diversity in IBD Patients

The strongest pattern identified in this study was a significant reduction in microbial diversity among IBD patients. Diversity indices including observed richness and Shannon diversity ranked among the most discriminative features, indicating that loss of microbial diversity is a key characteristic of IBD-associated dysbiosis.

8.2 Alteration of Beneficial Gut Microbial Taxa

Several beneficial microbial taxa associated with gut health emerged as important discriminatory biomarkers. The reduced abundance of these organisms in IBD patients suggests disruption of intestinal homeostasis and loss of protective microbial functions.

8.3 Dysbiosis as the Dominant Signature of IBD

The selected biomarkers indicate that IBD is characterized by a community-wide imbalance in microbial composition rather than the presence of a single pathogenic organism. Multiple beneficial taxa were altered simultaneously, suggesting widespread microbial dysbiosis. This finding aligns with current microbiome research indicating that disruption of the normal microbial ecosystem is a key feature of IBD pathogenesis.

8.4 Machine Learning Successfully Classified IBD

The high predictive performance achieved by all machine learning models demonstrates that gut microbiome composition contains sufficient biological information to reliably distinguish healthy individuals from IBD patients.

8.5 Significance of the Selected Biomarkers

- observed_richness: Interpretation: lower richness = reduced microbiome complexity
- shannon_diversity: Interpretation: lower values indicate ecological imbalance
- Akkermansia muciniphila: Interpretation: reduction may contribute to intestinal inflammation
- Bacteroides fragilis: Interpretation: altered abundance associated with inflammatory disorders
- Gemmiger formicilis: Interpretation: depletion may contribute to disease progression

9. Conclusion

This study successfully identified gut microbiome signatures capable of distinguishing healthy individuals from IBD patients using machine learning techniques.

Feature selection revealed that both microbial diversity metrics and specific microbial taxa played major roles in disease classification. Among the most discriminative biomarkers were:

- Gemmiger formicilis
- observed_richness
- shannon_diversity
- Alistipes shahii
- Barnesiella intestinihominis
- Akkermansia muciniphila
- Bacteroides fragilis

Among all evaluated models:

- XGBoost achieved the highest overall performance (Accuracy = 94.20%, AUC = 0.9803).
- Random Forest demonstrated excellent interpretability and strong classification accuracy.
- SVM achieved the highest recall (0.8571), indicating strong sensitivity for detecting healthy samples.
- Neural Network showed comparatively lower performance after feature reduction.

The findings support the hypothesis that gut microbial diversity and specific microbial taxa are strongly associated with IBD status. These microbial biomarkers have potential applications in non-invasive disease screening, microbiome-based diagnostics, and personalized therapeutic interventions for inflammatory bowel disease.